$y_i = b_0 + b_1 z_{1i} + b_2 z_{i2}$

1. $SSE = \sum_{i=1}^{N} (y_1 - b_0 - b_{1z_i} - b_{2i})^2$

2. $\frac{\partial SSE}{\partial b_0} = -2 \sum (y_1 - b_0 - b_1 z_{i1} - b_{z_{i2}}) = 0$

$\frac{\partial SSE}{\partial b_1} = -2 \sum z_{i1} (y_1 - b_0 - b_1 z_{i1} - b_{z_{i2}}) = 0$

$\frac{\partial SSE}{\partial b_2} \quad -2 \sum z_{i2} (y_1 - b_0 - b_1 z_{i1} - b_{z_{i2}}) = 0$

3. $e \cdot z = 0 \qquad \sum e = 0$

$e \cdot z_j = 0$ orthogonal

4. $b_0^* = \bar{y}$

$\sum y_1 = N b_0 + b_1 \cancel{\sum z_{i1}}^{0} + b_2 \cancel{\sum z_{i2}}^{0}$

$b_0 = \bar{y}$

$\sum z_{i1} (y_i - \bar{y}) = b_1 \sum z_{i1}^2 + b_2 \sum z_{i1} z_{i2}$

$\sum z_{i2} (y_1 - \bar{y}) = b_1 \sum z_{i1} z_{i2} + b_2 \sum z_{i2}^2$

5. $\begin{pmatrix} \sum z_{i1}^2 & \sum z_{i1} z_{i2} \\ \sum z_{i1} z_{i2} & \sum z_{i2}^2 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} \sum z_{i1} (y_i - \bar{y}) \\ \sum z_{i2} (y_i - \bar{y}) \end{pmatrix}$

6. $z_{ij} = x_{ij} - m_j$

$A = \begin{pmatrix} Var(x_i) & Cov(x_1, x_2) \\ Cov(x_1, x_2) & Var(x_2) \end{pmatrix} \qquad C = \begin{pmatrix} Cov(x_1, y) \\ Cov(x_2, y) \end{pmatrix}$

A is covariance matrix and C is vector.

Shows why multiple regression controls other variables